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Title: AI-Powered Risk Assessment in Financial Auditing: A Machine Learning Approach

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Summary:

This paper examines the application of machine learning (ML) techniques in enhancing risk assessment within financial auditing, focusing on how AI can improve the identification and prioritization of audit risks. The authors argue that traditional risk assessment methods, often reliant on manual processes and subjective judgment, are increasingly insufficient for handling the complexity of modern financial data. By leveraging ML algorithms, such as random forests and gradient boosting, the study demonstrates how AI can analyze historical financial data, detect high-risk areas, and predict potential misstatements with greater precision.

The methodology combines a quantitative evaluation of ML models on financial datasets from 25 multinational corporations (2017-2022) with qualitative feedback from audit professionals. The results show that ML-based risk assessment models achieve a 20-30% improvement in identifying high-risk transactions compared to conventional methods, with random forests outperforming others in handling noisy data. A case study of a tech firm illustrates how AI flagged understated liabilities missed by manual audits, reducing restatement risks by 15%. Challenges include the need for robust training datasets, computational resource demands, and ensuring model interpretability for regulatory acceptance.

The findings highlight that AI-driven risk assessment not only enhances accuracy but also cuts audit planning time by up to 22%, enabling auditors to allocate resources more effectively. However, the authors note barriers such as data privacy concerns and the need for auditors to develop technical skills to interact with AI systems. They advocate for integrating AI tools into existing audit frameworks and fostering collaboration between data scientists and auditors to ensure practical implementation. The paper underscores AI's potential to transform risk assessment into a more proactive and data-driven process, while calling for further research on ethical and regulatory implications.